Right Triangles & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Proving triangles congruent

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

2 [Foundational] • Target: Proving triangles congruent

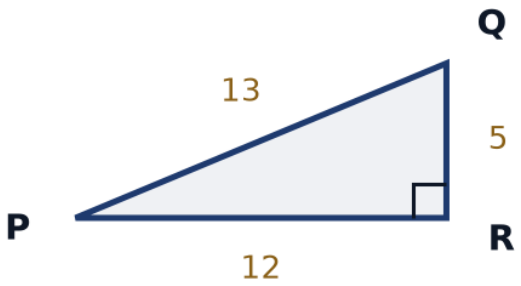
On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

3 [Foundational] • Target: Proving triangles congruent

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Proving triangles congruent

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.



5 [Foundational] • Target: Proving triangles congruent

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

6 [Foundational] • Target: Proving triangles congruent

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Proving triangles congruent

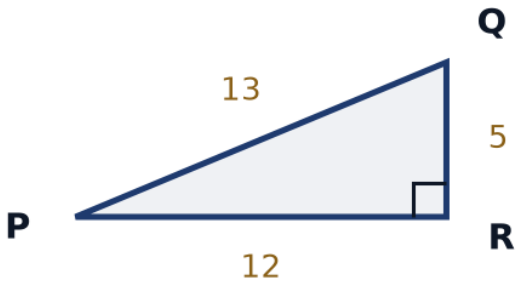
A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

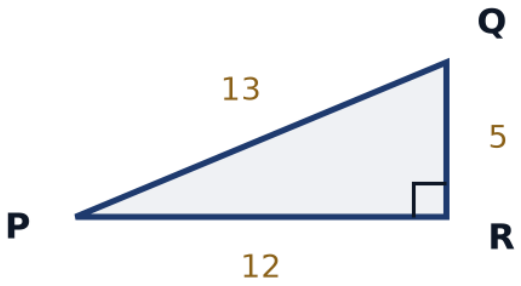
8 **[Intermediate]** • **Target: Proving triangles congruent**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



9 **[Intermediate]** • **Target: Proving triangles congruent**

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 **[Intermediate]** • **Target: Proving triangles congruent**

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

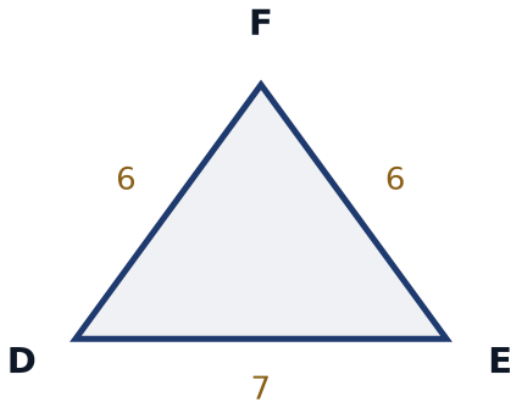
11 **[Intermediate]** • **Target: Proving triangles congruent**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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12 [Intermediate] • Target: Corresponding parts of congruent triangles

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: Corresponding parts of congruent triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: Corresponding parts of congruent triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

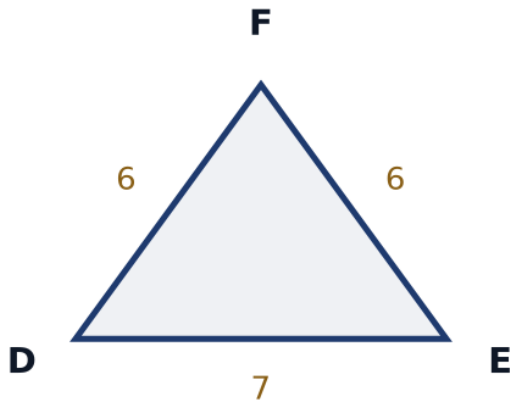
15 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

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16 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 10.01 Coordinate Geometry

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.



17 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 11.01 Probability

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

18 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 2.01 Lines & Angles

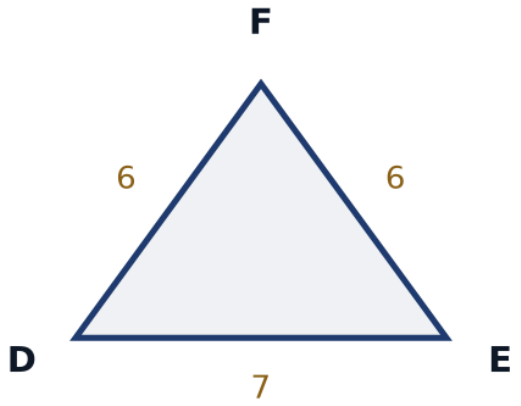
In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

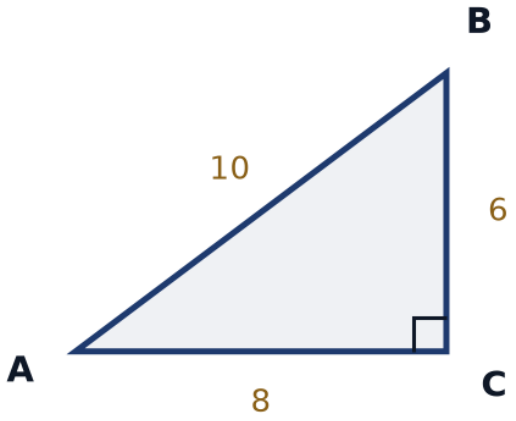


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Angles of triangles

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 81°

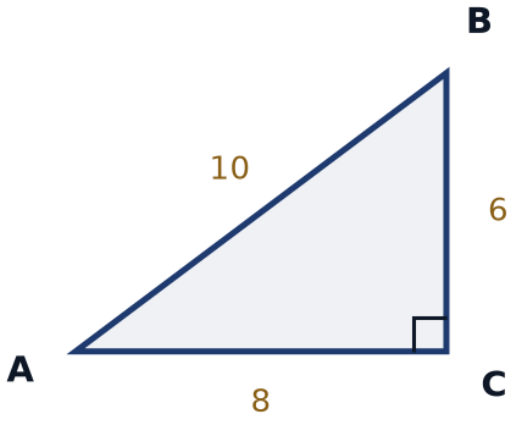
b. 91°

c. 73°

d. 126°

22 [Foundational] • Target: Angles of triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?



a. 21 cm

b. 24 cm

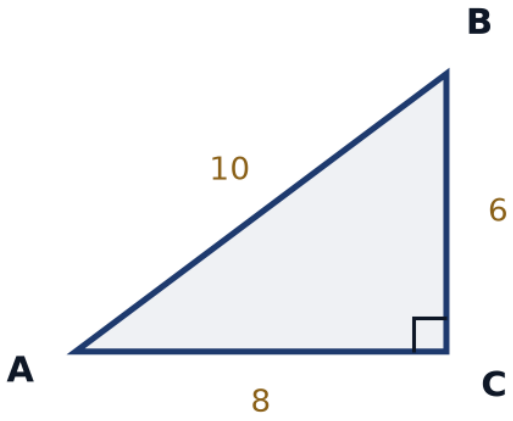
c. 32 cm

d. 72 cm

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23 [Foundational] • Target: Angles of triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?



a. 10

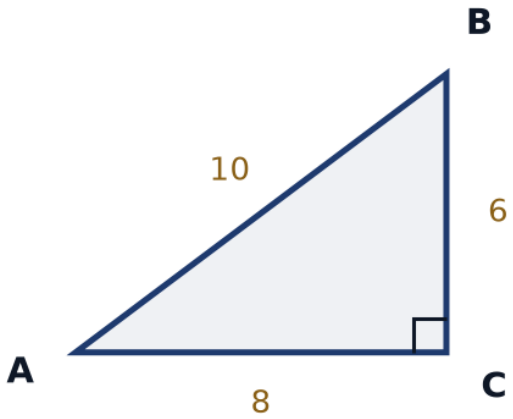
b. 14

c. 12

d. 14

24 [Foundational] • Target: Angles of triangles

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 60°

b. 70°

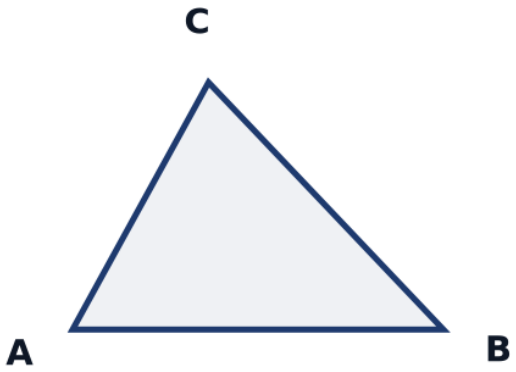
c. 52°

d. 126°

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25 [Foundational] • Target: Review: Identifying congruent triangles

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 105°

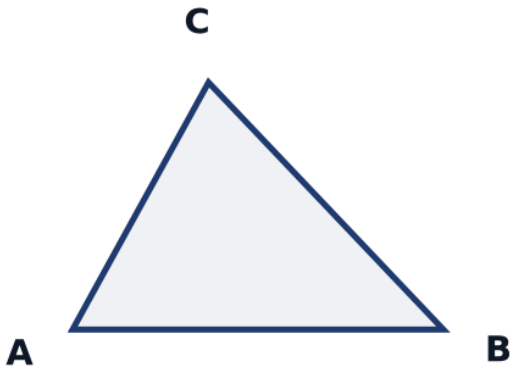
b. 115°

c. 97°

d. 150°

26 [Foundational] • Target: Review: Identifying congruent triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?



a. 9 cm

b. 12 cm

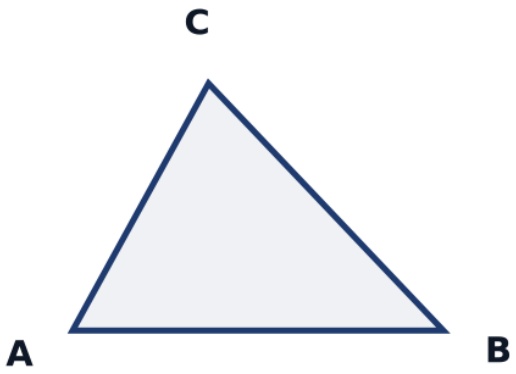
c. 16 cm

d. 36 cm

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27 [Foundational] • Target: Review: Identifying congruent triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?



a. 13

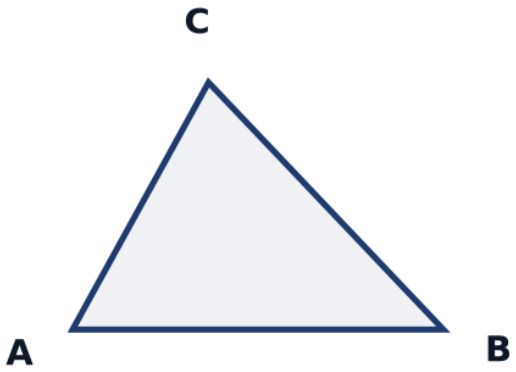
b. 18

c. 16

d. 19

28 [Foundational] • Target: Review: Identifying congruent triangles

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 84°

b. 94°

c. 76°

d. 150°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Angles of triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

30 [Intermediate] • Target: Angles of triangles

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 74°

b. 84°

c. 66°

d. 126°

31 [Intermediate] • Target: Angles of triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

32 [Intermediate] • Target: Angles of triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 12

b. 16

c. 14

d. 16

33 [Intermediate] • Target: Review: Identifying congruent triangles

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 98°

b. 108°

c. 90°

d. 150°

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34 [Intermediate] • Target: Review: Identifying congruent triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

35 [Intermediate] • Target: Review: Identifying congruent triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 16

b. 21

c. 19

d. 22

36 [Intermediate] • Target: Proving triangles congruent

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97°

b. 107°

c. 89°

d. 149°

37 [Intermediate] • Target: Proving triangles congruent

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Proving triangles congruent • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24

d. 28

39 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 3.01 Triangles & Congruence

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96°

b. 106°

c. 88°

d. 148°

40 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

41 [Challenging] • Target: Corresponding parts of congruent triangles • Cross-Lesson: 6.01 Polygons & Quadrilaterals

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29

d. 34

42 [Challenging] • Target: Isosceles and equilateral triangles • Cross-Lesson: 7.01 Circles

In ΔABC studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 147°

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43 [Challenging] • Target: Isosceles and equilateral triangles • Cross-Lesson: 8.01 Area, Surface Area & Volume

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

44 [Challenging] • Target: Isosceles and equilateral triangles • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 9

b. 13

c. 11

d. 13

45 [Challenging] • Target: Coordinate methods and triangle congruence • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 7

b. 12

c. 10

d. 13

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